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DISCOUNTING: A PRACTICAL GUIDE TO MULTILEVEL ANALYSIS OF CHOICE DATA

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Multilevel modeling provides the ability to simultaneously evaluate the discounting of individuals and groups by examining choices between smaller sooner and larger later rewards. A multilevel logistic regression approach is advocated in which sensitivity to relative reward magnitude and relative delay are considered as separate contributors to choice. Examples of how to fit choice data using multilevel logistic models are provided to help researchers in the adoption of these methods.

Key words: delay discounting, choice, multilevel modeling, statistical analysis

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Psychologists in all domains have been interested in the factors that drive a subject's choice between two or more options that vary along one or more dimensions. In the study of delay (or temporal) discounting, alternatives differ in the magnitude of their outcome and the delay to their receipt (Loewenstein, 1988; Rachlin, Raineri, & Cross, 1991). Subjects are offered a series of choices between smaller sooner (SS) rewards and larger later (LL) ones. By examining each subject's choices, researchers can identify individual and group differences in the degree to which a reward's magnitude is discounted as a function of its delay (see Odum, 2011, for a description).

Choice data provide a special challenge to estimating the rate of discounting and the shape of the discounting function. Three approaches have been commonly used. First, the individual choices can be used to identify a series of indifference points that are then subjected to curve fitting (e.g., Myerson & Green, 1995; Rachlin et al., 1991). For example, if a subject is equally likely to choose an option leading to two pellets now versus an option leading to eight pellets in 10 s, then eight pellets is assumed to have been discounted by 75% after a 10 s delay. A series of such choices at different delays is then used to identify the relationship between delay and discounted value that can then be fitted with possible discounting functions (Doyle, 2013). The most popular discounting function is the hyperbolic (Ainslie & Herrnstein, 1981; Chung & Herrnstein, 1967; Mazur, 1987; Rachlin et al., 1991):

$$Discounted\ Amount = \frac{Amount}{1 + k \times Delay} \tag{1}$$

The best fitting value of k for an individual or group represents the rate at which amounts are discounted as a function of delay.

One of the principal challenges to this approach is the need for indifference points, so various titration procedures have been used to derive them (e.g., Baker, Johnson, & Bickel, 2003; Kirby & Herrnstein, 1995; Madden, Petry, Badger, & Bickel, 1997). Calculating k (using Eq. (1)), Area Under the Curve (AUC; Myerson, Green, & Warusawitharana, 2001), and ED50 or $1/k$ (Yoon & Higgins, 2008) requires the presence of indifference points at which there is subjective equivalence of the SS and LL outcomes. Indifference points do not exist, however, for subjects or conditions in which discounting is not observed over the range of delays tested (e.g., Lane, Cherek, Pietras, & Tcheremissine, 2003; Young & McCoy, 2015) or for preparations that do not titrate to indifference (e.g., Evenden & Ryan, 1996). Although a k of zero and AUC of 1.0 could be inferred in the absence of discounting, these situations translate into an ED50 of infinity. Notably, Johnson and Bickel's (2008) second criterion suggests

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the exclusion of subjects for which discounting is too low, a procedure that can create biased discounting estimates. Even when a titration method is used, the production of indifference points requires a series of choices that must be aggregated. Aggregating assumes that the discounting rate is constant across trials and sessions, but impulsive choice has not been shown to be invariant to the method used (Peterson, Hill, & Kirkpatrick, 2015).

Second, a researcher could generate a set of profiles of choices consistent with an assumed discounting function for typical parameter values for that function (Gray, Amlung, Palmer, & MacKillop, 2016; Kirby & Marakovic, 1996; Kirby, Petry, & Bickel, 1999). This approach presents three difficulties, (a) for any particular function there is a range of parameter values consistent with a finite number of choices, (b) the parameter space must be adequately sampled in the creation of choice profiles, which is more problematic when more than one parameter must be estimated, (c) subject choice inconsistencies mean that an analyst must find the idealized choice profile that is most similar to a subject's choice profile or exclude a subject for which behavior is too inconsistent.

Third, a logistic regression of choice data can be used to indirectly estimate the discounting rate, k, for the hyperbolic discounting function (Wileyto, Audrain-McGovern, Epstein, & Lerman, 2004). This approach uses two predictors of choice in the following logistic regression equation (this equation is only applicable when the SS delay is 0):

$$P(waiting) = logistic\left(\beta_{magnitude}\left(1-\frac{LL_{magnitude}}{SS_{magnitude}}\right) + \beta_{delay}LL_{delay}\right) \quad (2)$$

where the two regression weights, $\beta_{magnitude}$ and β_{delay}, estimate the relative impact of the ratio of the LL and SS outcome magnitudes and that of the length of the delay to the LL outcome, respectively. Note that there is no intercept estimated (i.e., β_0 is zero). Wileyto et al. (2004) showed that the ratio, $\beta_{delay}/\beta_{magnitude}$, estimates the discounting rate, k, of the hyperbolic discounting function.

Each of these approaches has routinely required estimating discounting parameters like k for each of the N individuals or multiple ks for each individual, if there is a within-subject condition manipulation. The resulting estimates of k are then used as the outcome variable in a second stage of analysis in which the relationship between the predictors and k is assessed (usually, the k values are log-transformed due to their highly skewed distribution; Baker et al., 2003; Rachlin et al., 1991).

Challenges that Arise with the Common Analytical Approaches to Choice Discounting Data

In addition to the specific challenges already noted, each of these approaches has used a two-stage approach in which individual discounting rates are first determined and later used in a subsequent analysis (also see Young, 2017). For example, k may be estimated for each subject in a group of people who use heroin and for each subject in a group of nonusers, and a subsequent t-test can be used to compare these two sets of k values. Two-stage approaches suffer from the problem that the second stage analysis does not have any information regarding the precision of the estimates provided by the first stage analysis. In an experiment, some subjects perform more consistently than others and some subjects may have missing data or simply less data due to apparatus failure during a session. Although differential weighting of these estimates can be used to compensate for this challenge (Landes, Pitcock, Yi, & Bickel, 2010; Sidik & Jonkman, 2005), the weighting approach has been largely supplanted by multilevel modeling which estimates both the individual and group estimates simultaneously (de Leeuw & Meijer, 2008).

Another shortcoming of the two-stage approach is that deriving a k for a subject requires aggregating across multiple choices. Thus, any trial-by-trial changes in the preference for the SS or LL as a session proceeds are not captured in the first stage and thus not available to the second stage. One method to compensate for this problem is to use slope estimates from individual fits as outcomes for the second analysis stage (e.g., Landes et al., 2010; Wileyto et al., 2004), but this method encounters the same problems as any two-stage approach—information about the precision of each slope is not passed on to the second stage.

Of the three common methods, logistic regression is the least common but provides a good basis on which to build because it does not require any aggregation before estimating the discounting rate. A logistic regression takes as input the predictors and the individual choices and produces not only an estimate of the discounting rate but also a theoretically sound method of estimating the uncertainty of its components, β_{delay} and $\beta_{magnitude}$, as well as moderators of those slopes. Thus, the remainder of this manuscript will focus on logistic regression as the basis for understanding and modeling delay discounting choice.

One Source of Delay Discounting Behavior versus Two

The logistic regression approach to delay discounting estimates the independent effects of two factors, the delay to each alternative outcome and the magnitude of each outcome, using the regression weights β_{delay} and $\beta_{magnitude}$. Previous research has noted that a preference for the SS could be driven by a strong preference for immediacy and/or relative indifference to the magnitude disparity (Kheramin et al., 2002; Locey & Dallery, 2009; Marshall, Smith, & Kirkpatrick, 2014). Numerous studies have examined the relative control by magnitude and delay in reward choice (e.g., Grace, 1995; Ito & Asaki, 1982; Logue, Pena-Correal, Rodriguez, & Kabela, 1986; Rodriguez & Logue, 1986). Because the choice between a smaller–sooner outcome and a larger–later outcome involves the tension between smaller and larger and between sooner and later, the preference for one outcome over the other will be affected by the behavioral impact of the smaller–larger difference and the sooner–later difference. A subject indifferent about delay but sensitive to magnitude will choose the LL outcome regardless of the delay. But, if a manipulation increased sensitivity to the difference in delays without altering sensitivity to magnitude, this same subject would require a larger difference in magnitudes to produce choices of the LL outcome.

The potential for differential weighting of delay differences and magnitude differences has led to studies in which they are independently estimated. Locey and Dallery (2009) experimentally evaluated whether nicotine was increasing delay sensitivity or decreasing magnitude sensitivity in a rat sample. They determined that nicotine decreased preference for three pellets over one pellet without any effect on delay sensitivity and noted that the standard hyperbolic model could not reconcile their findings. Marshall et al. (2014) trained rats to make more accurate judgments of magnitude or delay and provided initial evidence that improved temporal discrimination produced more self-controlled choices. Thus, it is clear that under some conditions there will be differential sensitivity to reward delay and reward magnitude that cannot be captured by estimating k.

Evidence has also suggested that delay and magnitude may be processed by different brain regions (Ballard & Knutson, 2009). Ballard and Knutson (2009) reported that activity in certain brain regions (e.g., nucleus accumbens) was correlated with future reward magnitude whereas activity in other regions (e.g., dorsolateral prefrontal cortex) was correlated with future reward delay. They concluded that more impulsive subjects (i.e., those with a high k) showed less neural sensitivity to the larger magnitudes of delayed rewards but greater neural sensitivity to the delays to those rewards.

As a result of reports like these, it is increasingly important that the effects of delay and magnitude can be independently estimated rather than collapsed into a single discounting rate like k. If empirical evidence suggests that the regression weights for these two factors are highly correlated across subjects or conditions, then a focus on a single discounting rate would be justified (at least some of the time). But, as these studies demonstrate, some interventions might have differential impact on delay and reward sensitivity.

Multilevel Logistic Regression of Choice Discounting Data

Multilevel modeling involves estimating a model's parameters (intercepts and slopes in regression) at multiple levels. For repeated measures analysis, these two levels are the individual and the group. Unlike the two-stage approach in which parameters are estimated at the individual level first, and these estimates are used to infer group characteristics, multilevel modeling estimates both the individual and group parameters simultaneously (Gelman & Hill, 2006; Pinheiro & Bates, 2004). This approach means that not only are the group parameter estimates influenced by individual estimates but also that the individual parameter estimates are influenced by the group estimates. Solving the two levels of estimation simultaneously rather than sequentially results in more stable estimates of individual behavioral patterns because the complete data set informs those estimates, not just the data for that individual.

In considering the use of the multilevel approach, it is important to recognize that statistical modeling produces educated guesses regarding parameter estimates. When the data from an individual are plentiful and consistent, those data dominate inferences about that individual's behavior. Furthermore, data from many such individuals provide a strong basis from which to infer the behavior of a group comprising these individuals. However, some subjects provide very noisy data, highly implausible data (e.g., preferring delayed outcomes over immediate outcomes that are equivalent in magnitude), or very limited data (e.g., by not completing sessions or surveys). When making inferences about group performance, it is reasonable to give these subjects less weight. Traditionally, these subjects are omitted from a data analysis and thus given a weight of zero (e.g., Johnson & Bickel, 2008). Instead, multilevel modeling down-weights the influence of these subjects based on maximum likelihood estimation, thus eliminating human judgment from the process. A researcher should, however, still eliminate subjects or specific data points for which there is clear evidence to support their omission (e.g., due to subject sickness or equipment failure).

Multilevel modeling has a secondary consequence: The parameter estimates at the subject level are affected by the performance of the group to which this subject belongs (this process is called *shrinkage*). This behavior can be disturbing for researchers interested in individual differences. The influence of the group estimate on the individual estimate, however, is minimal when the subject has rich, consistent data, which is frequently the case in a choice experiment (this situation does not hold when fitting indifference functions which are usually based on a limited number of indifference points). It is only when the subject has inadequate, inconsistent, and/or unusual data that discernible shrinkage will be observed. Such data represent a poor basis for statistical or human inferences about the behavior of such an individual; multilevel modeling merely uses the data from the other subjects to make a more educated guess about this individual rather than simply dropping the individual from an experiment and thus completely eliminating their influence. Furthermore, dropping subjects can produce differential attrition, which has its own inferential consequences.

Because multilevel modeling uses maximum likelihood estimation rather than error minimization, it can elegantly handle design imbalance and missing values (Everitt, 1998; Gueorguieva & Krystal, 2004) either of which can prevent a data analyst from fitting data at the level of the individual trial. *Generalized* multilevel modeling extends the classic regression approach that assumes that the residuals are normally distributed to situations in which other distributions can be specified (e.g., binomial distributions for binomial data with a logit link function, and Poisson distributions for count data with a log link function). In sum, a repeated measures design in which the outcome is binomial (like the choice between SS and LL) can now be fitted with multilevel logistic regression using tools like PROC MIXED in SAS, MIXED in SPSS, fitglme in the Matlab Statistical Toolbox, and glmer in the free open source statistical language *R*. Because of its universal accessibility, the Supplemental Materials use the glmer command in the lme4 library of *R* (for related approaches using Bayesian multilevel modeling see Chavez, Villalobos, Baroja, & Bouzas, 2017; Vincent, 2016).

Simulated Data and Parameter Estimation

Logistic regression does not estimate the discounting rate, *k*, directly. Rather, it estimates the relative sensitivity to reward magnitude and delay by estimating the regression weight associated with each of these factors. Depending on the configuration of the logistic model, the ratio of these two regression weights can be used to directly estimate *k*. As Wileyto et al. (2004) demonstrated, *k* increases when the sensitivity to delay increases but *k* can also increase when the sensitivity to magnitude decreases. Thus, many different combinations of the regression weights β_{delay} and $\beta_{magnitude}$ can produce the same value of *k*. To estimate the differential effects of delay and magnitude requires a design in which these two factors are manipulated independently.

The most commonly used method for assessing delay discounting in humans is the Kirby et al. (1999) delay discounting questionnaire which produces a series of 27 SS/LL choices for each subject. Unfortunately, in this questionnaire the particular combinations of reward ratios and delay durations are highly correlated (see Fig. 1) which makes it effectively impossible to clearly distinguish their separate effects due to multicollinearity. Thus, a researcher interested in independently assessing sensitivity to reward magnitude and delay will need to design a study to ensure limited or no collinearity. In a set of simulations, I will combine various levels of magnitudes and delays to more fully sample the stimulus space, thus allowing for the separate estimation of the two regression weights.

A key issue when advocating a different approach to data analysis is to evaluate its performance relative to alternative approaches. To answer that question, scientists use Monte Carlo simulation in which data with known population properties are created, and multiple techniques are applied to those data to determine their relative ability to recover the population characteristics. The current section, therefore, presents some limited simulations to document the performance of a two-stage approach (in which choices are used to estimate individual subject k values or to estimate individual β_{delay} and $\beta_{magnitude}$ weights, each of which can be analyzed in a second stage) and the multilevel logistic regression approach.

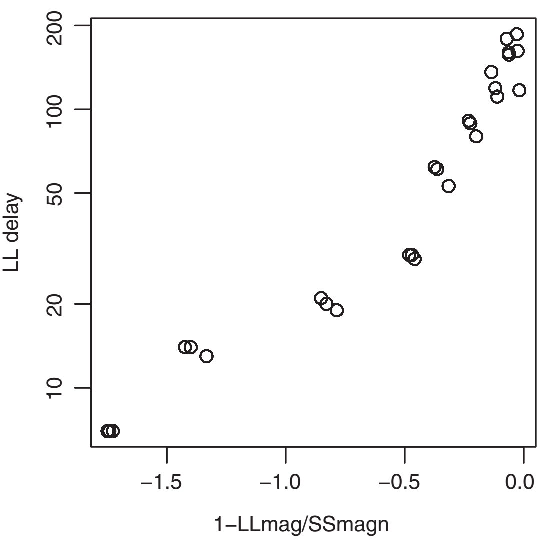

Fig. 1. The relationship between the two predictors used in the Wileyto et al. (2004) logistic regression model in the Kirby et al. (1999) survey design.

The data set tested was simulated using a design in which the SS outcome was always the same (a unit magnitude outcome at a zero delay) and the LL outcome was delayed either 1, 2, 4, 8, or 16 time units and was either a 1, 2, 4, or 8-unit magnitude. There were 5 × 4 or 20 trial types presented in each of five blocks, thus producing 100 choices for each simulated participant. Each of the 50 participants was simulated to have a different sensitivity to delay and magnitude (see left side of Fig. 2). For the five blocks of 20 trials, the participant's choice of the LL outcome was determined by choosing a random binomial outcome with the following probability:

$$P(LL) = logistic\left(\beta_{magnitude} * ln\left(\frac{LLmagnitude}{SSmagnitude}\right) - \beta_{delay} * ln(LLdelay + 1)\right) \quad (3)$$

Thus, when the magnitudes were identical or the LL delay was zero (and thus equal to the SS delay), the logarithmic terms reduce to $ln(1)$ or zero, resulting in the respective term having no impact on the probability of choosing the LL outcome. The simulated data set is available on the author's website at http://www.k-state.edu/psych/research/young/suppmaterial.html. A logarithmic relationship was used because prior research has suggested that hyperbolic-like behavior is exhibited by subjects who show a logarithmic sensitivity to time duration (Cui, 2011; Ebert & Prelec, 2007; Zauberman, Kim, Malkoc, & Bettman, 2009), and log-sensitivity to both magnitude and duration is consistent with prior research in psychophysics (Fechner, 1860).

The design used to generate the current data set precludes a standard indifference point analysis because there was no attempt to ensure that for each delay, there was a magnitude that created indifference, *P(SS)* = *P (LL)* = .50. However, individual k values could be derived by fitting the Wileyto et al. (2004) equation (Eq. (2)) to each subject; the

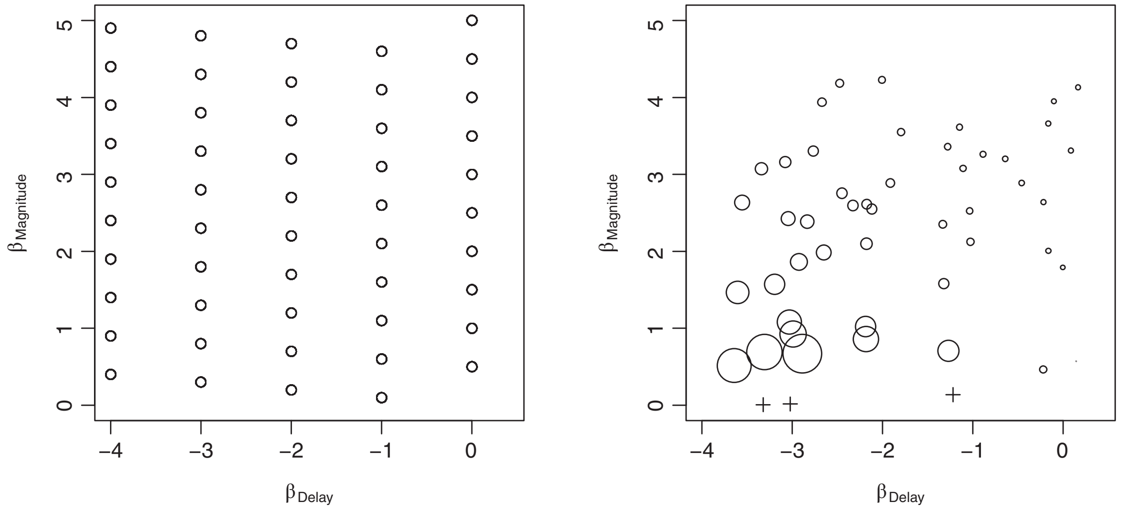

Fig. 2. (Left) Design of simulations. Each point represents a simulated subject's two population weights used in Equation (3) to generate the data. (Right) Result of multilevel logistic regression analysis of this data. Each point represents a simulated subject's two regression weights derived from a multilevel logistic regression. The size of each point corresponds to the *k* value derived from applying Equation (2) to each individual (three subjects designated with a cross did not generate reasonable *k* values). The correspondence between the left and right figures is illustrated in the bottom two graphs of Figure 3.

resulting discounting rates were then compared to the regression weights derived from a multilevel logistic model based on Equation (3) as well as used as the basis for a two-stage group comparison analysis. Because Equation (3) was used to generate the data, it was used as the gold standard. Equation (2) was only used to derive an estimate of *k*.

Individual fits of Equation (2) produced a median k ($\beta_{delay}/\beta_{magnitude}$) of 0.43, and individual subject estimates ranged from a minimum *k* of -14.60 to a maximum of 28.90. A subset of 6 of the 50 subjects generated negative *k* values with three of these being very close to zero. The distribution of *k* values was very skewed (which is frequently observed) and thus subjected to a log transformation, a common approach when analyzing discounting rates (e.g., Baker et al., 2003). This procedure was problematic in the face of negative *k* estimates, so 1.0 was added to each estimated *k* before log transformation. Even after this adjustment, three subjects still had *k*s well below 0 and thus produced missing values after transformation. The remaining 47 *log(k)* values were used in subsequent analyses.

Using a multilevel logistic regression approach, Equation (3) was fitted to the individual choices of all 50 subjects simultaneously (see the *R* code in the Supplemental Materials for details). The multilevel model and all subsequent models allowed both of these regression weights to vary randomly across subjects (in multilevel model parlance, these two slopes were included in two Subject × Predictor random effects; for details, see the Supplemental Materials). The analysis estimated separate regression weights for each choice attribute, β_{delay} = -1.93 (SE = .20, z = -9.72, p < .001) and $\beta_{magnitude}$ = 2.32 (SE = .21, z = 11.06, p < .001). As shown in the left side of Figure 2, the simulated subjects had a range of 0.0 to -4.0 for β_{delay} and a range of 0.1 to 5.0 for $\beta_{magnitude}$, thus the observed estimates were consistent with the average subject having population regression weights of β_{delay} = -2.0 and $\beta_{magnitude}$ = 2.5. A multilevel regression also produces estimates for each subject; these values were used to compare the performance of the two analyses.

The right side of Figure 2 plots each subject's estimated magnitude regression weight as a function of its estimated delay regression weight. The distribution of values is consistent with the range of values from which the choices were derived (see left side of Fig. 2). Importantly, the size of each dot corresponds to the *k* estimate derived from individual fits of Equation (2). This figure thus demonstrates that similar k values can be produced by very different sensitivities to reward delay and magnitude.

A Direct Comparison of Individually-Derived and Multilevel Regression Weights

The first analysis did not allow a direct comparison of the regression weights derived from individual regressions (based on Eq. (2)) and those from a multilevel regression (based on Eq. (3)) to the weights used to produce the simulated data (which used Eq. (3)). This shortcoming was by design because the individual fits used the Wileyto et al. (2004) model to produce estimated k values. In the next analysis, each individual was fitted using logistic regressions based on Equation (3) to allow direct comparison of a two-stage approach to the multilevel logistic regression approach.

The first result of note was that the individual logistic regression for Subject 50 produced perfect separation. Perfect separation in logistic regression occurs when the two classes of behavior (in this case, choosing SS vs. LL) can be perfectly predicted using a provided predictor; separation means that there is insufficient constraint on the parameter estimates (Albert & Anderson, 1984). Subject 50 produced an estimated $\beta_{magnitude}$ of 28.9 with a SE of 4398.8, clear evidence of a high level of parameter uncertainty when estimating its value based only on this subject's behavior. Perfect separation is much more likely to occur when fitting a limited number of individual choices than when fitting a multilevel model because the latter analysis is based on every subject.

Figure 3 illustrates the relationship between the estimated regression weights from the two types of analyses and the actual value that produced the simulated data. It is important to note that the original data contained noise so that the data provide an uncertain basis for estimating the population regression weight, a situation that would exist when analyzing real data. Both analyses provide good estimates of each individual's population regression weight, although the discrepant Subject 50 was not included in the plot because its outlier status dominated the plot. For the most extreme population regression weights ($\beta_{magnitude}$ > 4.0 and β_{delay} of -4.0), the multilevel approach exhibited some shrinkage such that the estimated weights were brought closer to the overall mean. Critically, the SEs of these weights indicated substantial estimation uncertainty in these parameters even in the absence of perfect separation. For example, Subject 48 produced a 95% CI for ($\beta_{magnitude}$ of [3.64, 8.63] and Subject 3 produced a 95% CI for β_{delay} of [-9.96, -2.18]; the multilevel estimates for these parameters exhibited shrinkage with estimates of 4.18 and -3.64, respectively, both of which were within the 95% CI of the individual fits. It is this high level of parameter uncertainty that produces shrinkage when data from subjects like these are fitted using a multilevel logistic regression.

Holding One Ratio Constant

It is common that one of the magnitude or delay ratios is held constant in a delay discounting experiment. This situation can be addressed by either including the variable as a predictor in the analysis so that the regression weight, $\beta_{magnitude}$ or β_{delay}, associated with this constant variable directly estimates its effect, or by using the intercept, β_0, to estimate the variable's impact. If the intercept is used, then the value of the constant ratio must be divided out to produce the same answer that the first method produces. For example, if the magnitude ratio was a constant 5:1, then β_0 in the second analysis is estimating $\beta_{magnitude} \times \log(5)$.

If held constant, the choice of the ratio is important to the proper estimation of both $\beta_{magnitude}$ and β_{delay}. If the ratio is chosen such that subject behavior is more extreme (i.e., nearly always choosing the SS or LL), then there is insufficient behavioral variation on which to estimate the regression weights. To demonstrate this effect, I revisited the simulated data set by fitting the multilevel logistic regression to five subsets of the data, each of which only involved one of the values of LL_{delay}. The results are shown in Figure 4. As the LL_{delay} grew larger, there was a greater incentive for the subjects to always choose the SS outcome, which both biased the β_{delay} estimate and greatly increased its uncertainty. In contrast, the broad sampling of the magnitude ratio helped to ensure the stability of its estimate at most values of the LL_{delay}.

Detecting Group Differences

Of central interest to someone studying delay discounting is the ability of an analytical

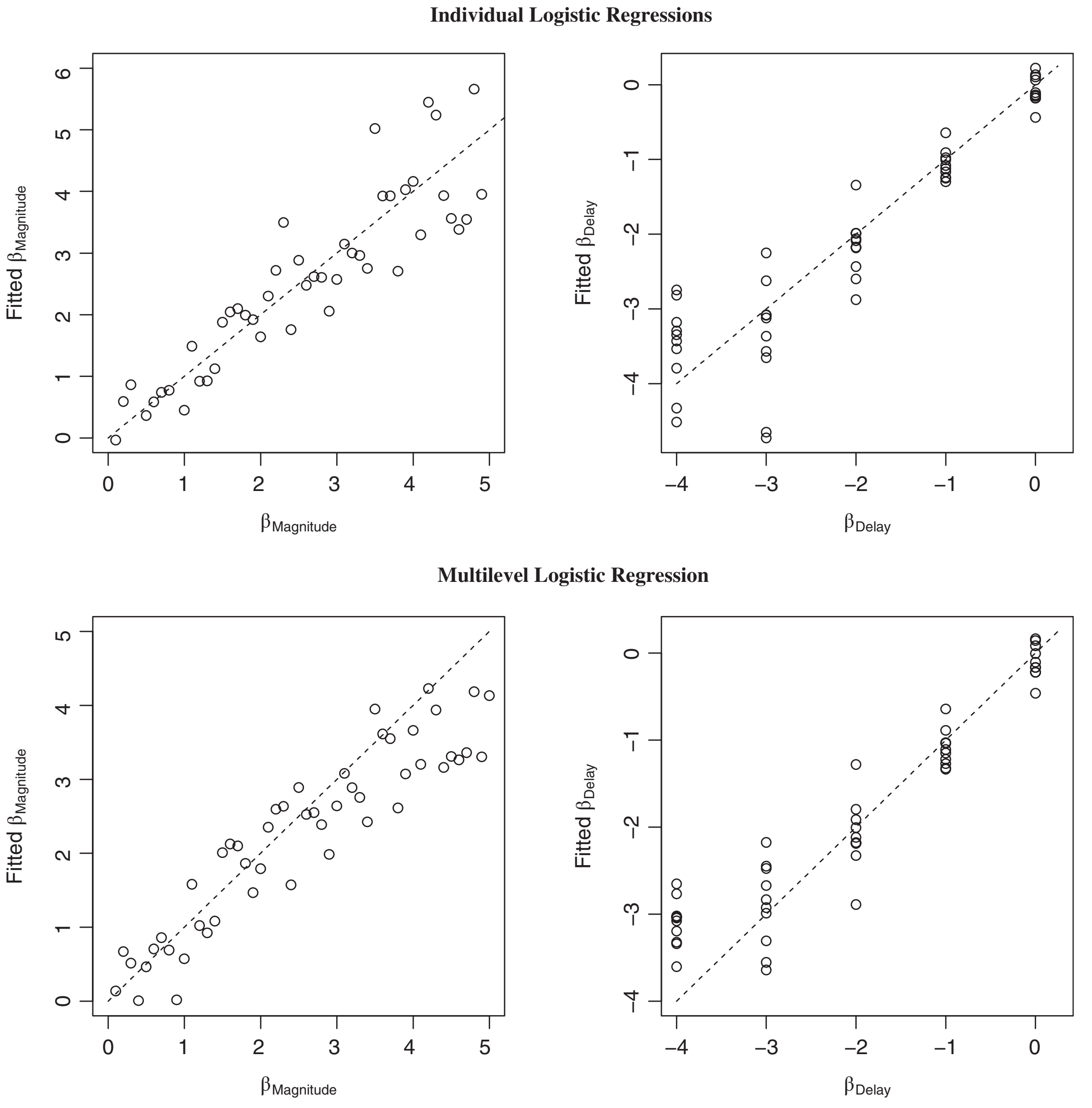

Fig. 3. (Top) Relationship between individually estimated fits of beta weights and population beta weights used to produce the data. One value for Fitted $\beta_{Magnitude}$ (estimate = 28.9) is omitted. (Bottom) Relationship between multilevel fits of beta weights and population beta weights. Each point represents a simulated subject.

approach to estimate the impact of independent variables of interest. To demonstrate the difference across approaches using a simple design, the data set used in the previous analysis was subjected to a test of group differences. The simulated subjects were divided based on their subject number (Group 1: 1-25 vs. Group 2: 26-50) because these subjects were simulated such that the sensitivity to magnitude increased linearly with subject number whereas sensitivity to delay was uncorrelated with the size of the subject number. Thus, the population values for these two groups have a different average sensitivity to magnitude but not to delay (mean $\beta_{magnitude}$ = 3.75 vs. 1.25, mean β_{delay} = -1.92 vs. -2.08 for Groups 1 and 2, respectively). A two-stage approach based on the k values derived earlier should identify that these two groups have different average ks, but the approach cannot determine whether the difference is

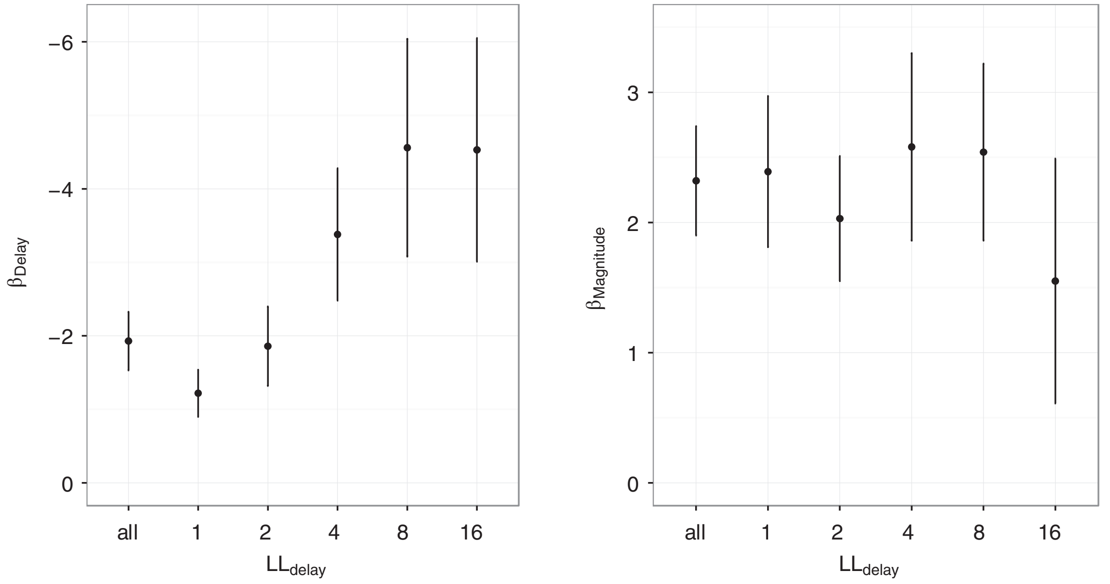

Fig. 4. The 95% confidence interval on each regression weight when a multilevel logistic regression was fitted to all of the data or to subsets of the data for which the LL_{delay} was held constant at each of the five values tested.

due to differences in delay sensitivity, magnitude sensitivity, or some combination of both. In contrast, the multilevel logistic approach can identify the overall group difference while also identifying that this difference is determined by differential sensitivity to magnitude, not delay. Furthermore, the two-stage approach may be less able to identify the group difference due to the reduced statistical power caused by data aggregation (i.e., information about each subject's sample size and its behavioral variability are not represented in the *k*s derived in the first analysis stage). In the following analyses, I will consider a two-stage approach based on the *k*-estimates derived from Equation (2), the multilevel logistic regression approach using Equation (3), and a final two-stage approach based on the delay and magnitude weights derived from Equation (3).

The second stage analysis of $\log(k)$ successfully identified that the subjects in Group 1 had lower discount rates than those in Group 2, $\log(k)$ = .33 and .86, respectively, $t(45) = 3.76$, $p < .001$. Recall that three subjects were excluded from the second stage of analysis due to their producing very negative *k* values that were not easily log-transformed; all three of these subjects were in Group 1 which raises concerns about differential attrition. As an aside, a *t*-test based on all 50 of the untransformed original *k* values was not significant, $t(48) = 1.84$, $p = .07$, nor was a nonparametric Wilcoxon rank sum test ($W = 220$, $p = .07$), but a permutation Wilcoxon test was significant ($Z = 2.60$, $p = .01$). The nonparametric Wilcoxon tests can only identify differences and do not estimate the *k* in each group. Note that the condition *k* distributions overlapped because *k* is based on both magnitude sensitivity (which did not overlap) and delay sensitivity (which overlapped significantly).

The multilevel logistic regression of group differences incorporates the independent variable of group into the original analysis and computes an estimate of the group differences in the regression weights. To simplify the presentation and to parallel Equation (2), the analysis excluded the intercept thus assuming that the estimated probability of choosing the larger later outcome is 0.5 when the SS and LL outcomes have the same delay and magnitude (i.e., the SS is neither sooner nor smaller than the LL). The regression thus estimated four parameters: $\beta_{magnitude}$, β_{delay}, $\beta_{Group \times magnitude}$, and $\beta_{Group \times delay}$. Given that group was dummy coded (0 = Group 1, 1 = Group 2), the latter two weights estimate the size of the difference in magnitude and delay sensitivity between the two groups. The results of the analysis are shown in the top of Table 1.

The two values of greatest interest in the top of Table 1 are those for the two interactions. The magnitude slope was 2.24 points lower in Group 2 (3.45 vs. 1.21, for Groups 1 and 2 respectively, z = -8.40; these values are similar to the population values of 3.75 and 1.25) whereas the delay slope was 0.22 points lower (here, this indicates an increase in sensitivity due to the difference in valence), a nonsignificant difference (-1.82 vs. -2.04, for Groups 1 and 2 respectively, z = -0.56; the values are similar to the population values of -1.92 and -2.08). The multilevel analysis thus correctly identified that the increase in discounting rate for Group 2 (documented in the two-stage k-based analysis) was due to a decrease in the behavioral control exhibited by magnitude and not due to an increase in control by delay. Furthermore, the multilevel analysis was much more powerful in identifying this group difference by producing a z-score of -8.40 versus the t-score of 3.76 produced by the two-stage k-based analysis (z and t magnitudes are comparable for the sample sizes used here).

In the final analysis, the individually-derived Equation (3) logistic regression weights from the earlier first-stage analysis were subjected to two second-stage analyses, one for $\beta_{magnitude}$ and one for β_{delay}. The results are shown in the bottom of Table 1. The group difference for $\beta_{magnitude}$ was detected but with serious misestimation of Group 1 (4.60 vs. 1.31 for Groups 1 and 2, respectively) caused by Subject 50's extreme value. Although a data analyst is unlikely to include this outlier, its exclusion creates a bias in the estimate by omitting the subject with the strongest sensitivity to magnitude while also artificially decreasing the estimated variance. Systematic attrition of extreme subjects is more likely for individually derived estimates because behavior that evidences very high sensitivity to magnitude or delay is more likely to create perfect separation. More germane to the current discussion is the fact that the multilevel analysis did not produce a problem with perfect separation and thus required no decisions regarding what to do with an outlier produced by this problem.

Applying Multilevel Modeling to Titration Data

Various titration methods have been proposed to estimate a subject's indifference point at each delay. One or more decision variables (e.g., the $SS_{magnitude}$ or LL_{delay}) are gradually changed until the subject is indifferent between the two options (e.g., Baker et al., 2003; Kirby & Herrnstein, 1995; Madden et al., 1997; Mazur, 1987; Richards, Mitchell, de Wit, & Seiden, 1997). Some of these methods thus produce different numbers of choices for each subject depending on how quickly the algorithm converges and will oversample options near the indifference point (e.g., there is usually little reason to explore mild variations on the choice between \$100 now versus \$1,000,000 in a week). In the next application of multilevel logistic regression, I analyzed data from a money discounting task obtained using a titration procedure (Jarmolowicz, Lemley, Asmussen, & Reed, 2015). Jarmolowicz et al. (2015) had subjects make decisions between a constant $LL_{magnitude}$

Table 1

Results of logistic regressions of group differences using a multilevel logistic regression (top) or two regressions of the weights derived from individual logistic regressions (bottom)

Predictor	β Estimate	SE	z/t	p
Multilevel Logistic Regression				
$\text{Log}(LL_{magnitude}/SS_{magnitude})$	3.45	0.21	16.78	< .001
Group × $\text{Log}(LL_{magnitude}/SS_{magnitude})$	-2.24	0.27	-8.40	< .001
$\text{Log}(LL_{delay} + 1)$	-1.82	0.27	-6.70	< .001
Group × $\text{Log}(LL_{delay} + 1)$	-0.22	0.39	-0.56	0.57
Individual Logistic Regressions				
$\text{Log}(LL_{magnitude}/SS_{magnitude})$	4.60	0.74	6.18	< .001
Group Effect for $\text{Log}(LL_{magnitude}/SS_{magnitude})$	-3.29	1.05	-3.13	.003
$\text{Log}(LL_{delay} + 1)$	-1.91	0.29	-6.66	< .001
Group Effect for $\text{Log}(LL_{delay} + 1)$	-0.27	0.40	-0.66	0.51

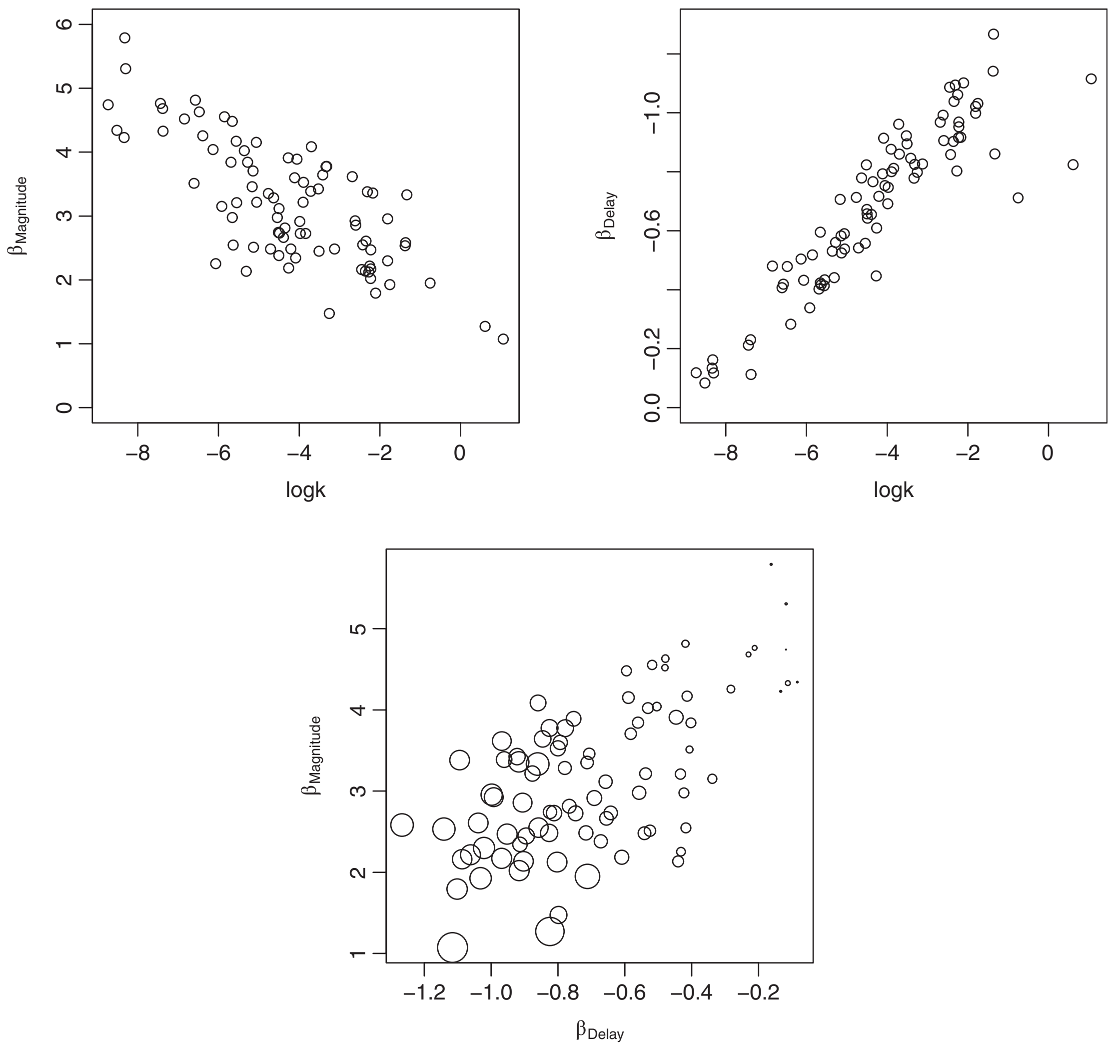

Fig. 5. (Top) Relationship between multilevel logistic regression estimates of beta weights and a multilevel nonlinear model estimate of *k* values derived from indifference points for Jarmolowicz et al. (2015). (Bottom) For the same data, the relationship between these estimated beta weights; dot size corresponds to each subject's *k* estimated from indifference points.

of \$10 at five LL_{delays} between 1 day and 1825 days (5 years) versus a constant SS_{delay} of 0 days while varying the $SS_{magnitude}$ between \$0 and \$10 to titrate the indifference point. These indifference points were used to produce *k* estimates by fitting a hyperbolic discounting function using multilevel modeling (Young, 2017), whereas the individual choices were subjected to a multilevel logistic regression based on Equation (3) but with the need to add a constant (1.0) to each magnitude because zero magnitudes were occasionally produced by the titration algorithm.

The results of the analysis are shown graphically in Figure 5. The data revealed stronger control by magnitude ($\beta_{magnitude}$ = 3.24, SE = 0.17) than by delay (β_{delay} = -0.70, SE = 0.04) with both regression weights significantly different from zero. The individual regression weights were strongly correlated with the log(*k*) (estimated using indifference points) in the direction expected (top of Fig. 5)—stronger control by magnitude was associated with a smaller *k* whereas stronger control by delay was associated with a higher *k*. Finally, the two regression weights were highly correlated such that subjects with stronger control by magnitude tended to show weaker control by delay (bottom of Fig. 5). The size of each dot in the bottom of Figure 5 corresponds to the *k* value estimated from the indifference analysis. As expected, subjects with the strongest control by delay and weakest control by magnitude (lower left corner of this figure) produced the highest discounting rates whereas those with the strongest control by magnitude and weakest control by delay (upper right corner) produced the lowest discounting rates. This analysis illustrates that titration data can be productively analyzed using the logistic approach.

To exercise the power of a multilevel logistic approach, the above analysis was supplemented by adding (centered) trial as a continuous predictor of the rate of change in magnitude and delay sensitivity thus addressing an issue not answerable with *k*-based analyses. The resulting model estimates four regression weights, $\beta_{magnitude}$, β_{delay}, $\beta_{Trial \times magnitude}$, and $\beta_{Trial \times delay}$. The interpretation of $\beta_{magnitude}$ and β_{delay} changes relative to the first analysis; the inclusion of a centered trial variable alters the analysis because subjects experienced different numbers of trials (varying from 27 to 218). For this new analysis, the regression weights at the mean trial of 36 were 4.02 (SE = 0.28) and -0.81 (SE = 0.05) for magnitude and delay, respectively. This analysis revealed a statistically significant Trial × log(magnitude ratio) interaction (z = 2.30, p = .02) but a nonsignificant Trial × log(delay ratio) interaction (z = 0.93, p = .35). Subjects demonstrated an increase in sensitivity to magnitude from Trial 1 to Trial 50 (3.31 to 4.28) whereas sensitivity to delay was unchanged (-0.86 to -0.79). Conclusions involving trial, however, should be treated with caution given that this titration procedure produced considerable variation in the number of trials experienced by each subject. Furthermore, the nature of those trials was not constant across time with later trials having less variation in SS_{delay} when the algorithm was converging on an indifference point.

Extension of the Logistic Approach to Designs Involving SS Delays That Are Not Zero

Although the bulk of research on delay discounting has involved a SS delay that is zero, some projects have presented participants with the choice between two delayed rewards (e.g., Calvert, Green, & Myerson, 2011; Green, Myerson, & Macaux, 2005; Mitchell & Wilson, 2012). Equations (2) and (3) do not apply when the SS delay is not zero. Extending the logistic regression approach to nonzero delays thus required the consideration of some variations on these basic models.

To analyze designs that include a nonzero SS delay, it is necessary to replace LL_{delay} with a delay ratio. The use of "LL_{delay} + 1" in Equation (3) was to foreshadow this necessity. Given that SS_{delay} is often zero, it is not possible to use LL_{delay}/SS_{delay} as the delay ratio. Although adding any positive constant achieves the goal of avoiding division by zero (Tukey, 1977), I chose a constant of 1.0 for simplicity. Future researchers can evaluate whether the choice of constant plays a significant role in fitting various data sets.

In the first demonstration of using logistic regression to fit choices between SS and LL outcomes with nonzero SS delays, I modified the earlier simulated data by having the SS delay be 0.00, 0.25, 0.50, 0.75, or 1.00. Thus, the 100 rows for each subject comprised a full factorial of 5 SS delays × 5 LL delays × 4 LL magnitudes. SS magnitude was held constant for convenience. Each subject's choices were again derived using Equation (3) with each subject having the same delay and magnitude slopes used in the first simulation; these data are also available on the author's website.

When fitting these data, the only modification to the multilevel logistic regression was the use of $\log((LL_{delay} + 1)/(SS_{delay} + 1))$ as the second predictor in the equation. The results were indistinguishable from those of the first simulation using a zero SS delay: β_{delay} = -1.94 (SE = .22, z = -8.84, p < .001) and $\beta_{magnitude}$ = 2.49 (SE = .21, z = 11.92, p < .001), again consistent with the average subject having population regression weights of β_{delay} = -2.0 and $\beta_{magnitude}$ = 2.5. The inclusion of a nonzero SS delay was easily incorporated into this logistic analysis.

To determine how this form of the logistic equation would generalize to real choice data, I fitted the model to Phase 1 data from the SYS-GE condition from Peterson et al. (2015). In this condition, rats were given the choice between a SS reward (one pellet at a 2 s delay) and a LL reward (two pellets at a 5, 15, 30, or 60 s delay). The LL delay was 5 s for the first five sessions, 15 s for the next five sessions, and so on, for a total of 20 sessions of training. Because the LL magnitude ratio was constant, the logistic regression included the intercept to estimate the magnitude effect (for a variation and further explanation, see the Supplemental Materials). After an initial fit, I also tested the data for changes in the parameter estimates across trials within each session and across the five sessions of training at each delay. Most of the rats completed all 54 trials in each session, but some did not; these unequal sample sizes were not an issue because multilevel modeling properly accommodates unbalanced data (Everitt, 1998).

The best fitting model that included only the intercept and delay ratio produced the expected positive intercept (9.63, SE = 0.76, z = 12.64) and negative delay slope (-4.22, SE = 0.34, z = -12.58). Even though the SS and LL magnitudes were constant throughout, the analysis can still estimate their impact on behavior; a rat that responds exclusively based on magnitude would show a strong positive intercept (i.e., a high base probability of choosing the LL outcome) and a delay slope near zero.

The model was then updated to include the possibility that the intercept (a proxy for bias produced by the magnitude ratio) and the delay slope (assessing sensitivity to the delay ratio) were changing across trials within a session. There are two challenges in interpreting the results of a logistic analysis involving an interaction: a) the main effect of each variable is contingent on the other variables having a value of 0, and b) the interaction is highly collinear with its main effects if the variables only comprise positive values. Thus, to simplify interpretation researchers will often center the predictors (subtract their means) and use effect coding of categorical variables. An analysis of variance uses effect coding for precisely this reason. For the following analysis, the trial predictor was centered but the log(delay ratio) was not because doing so would change the interpretation of the intercept. Recall that a properly counterbalanced design should produce a *P(LL)* of 0.5 when the magnitude and delay ratios are both 1, so deviations from a *P (LL)* of 0.5 indicate an effect due to the factor that was held constant (in this case, the magnitude ratio). Centering the log(delay ratio) would violate this condition. The results are shown in Table 2.

Table 2

Model fit for a multilevel logistic regression to the Peterson et al. (2015) data set

Predictor	Estimate	SE	z	p
Intercept	9.91	0.79	12.46	< .001
Trial	0.06	0.01	6.38	< .001
$\text{Log}((\text{LL}_{\text{delay}} + 1)/(\text{SS}_{\text{delay}} + 1))$	-4.36	0.35	-12.51	< .001
$\text{Trial} \times \text{Log}((\text{LL}_{\text{delay}} + 1)/(\text{SS}_{\text{delay}} + 1))$	-0.035	-0.004	-8.88	< .001

Note: The trial predictor was mean centered; the average trial was 27.44. Because $\text{LL}_{\text{magnitude}}/\text{SS}_{\text{magnitude}}$ was held constant, the intercept assesses the effect of the constant magnitude ratio on *P(LL)* at a delay ratio of 1.0, and the Trial main effect assesses the changes in this magnitude sensitivity at the same ratio (both weights would need to be divided by log(2) to estimate $\beta_{magnitude}$). The log(delay ratio) assesses the changes in *P(LL)* as the delay ratio varied, and the Trial × log(delay ratio) interaction assesses changes in this sensitivity.

The trial main effect (change in intercept across trials indicating a change in magnitude sensitivity) and the Trial × Delay Ratio (change in delay slope across trials) were both significant. The effects are depicted in Figure 6. The intercept (i.e., magnitude effect) averaged 9.91 (corresponding to a *P(LL)* of 99.9%) across the session when the delay ratio was 1.0, and increased from 8.24 at the beginning of each session to 11.57 at the end of each session. In contrast, the sensitivity to delay averaged -4.36 across the session, but grew stronger from the beginning (slope = -3.44) to the end of each session (slope = -5.28). A deeper examination of the behavior of this model revealed that the effect of magnitude actually decreased across trials at longer delays due to the growth in the behavioral control by delay eclipsing the growth in magnitude. For model details, see the Supplemental Materials.

The use of the delay ratio when nonzero delays are present worked well for this data set and demonstrated how the approach could be extended to a choice design involving nonzero delays and prolonged and blocked exposure to various delays with a constant magnitude ratio; this analysis revealed that sensitivity to delay increased within a session whereas the

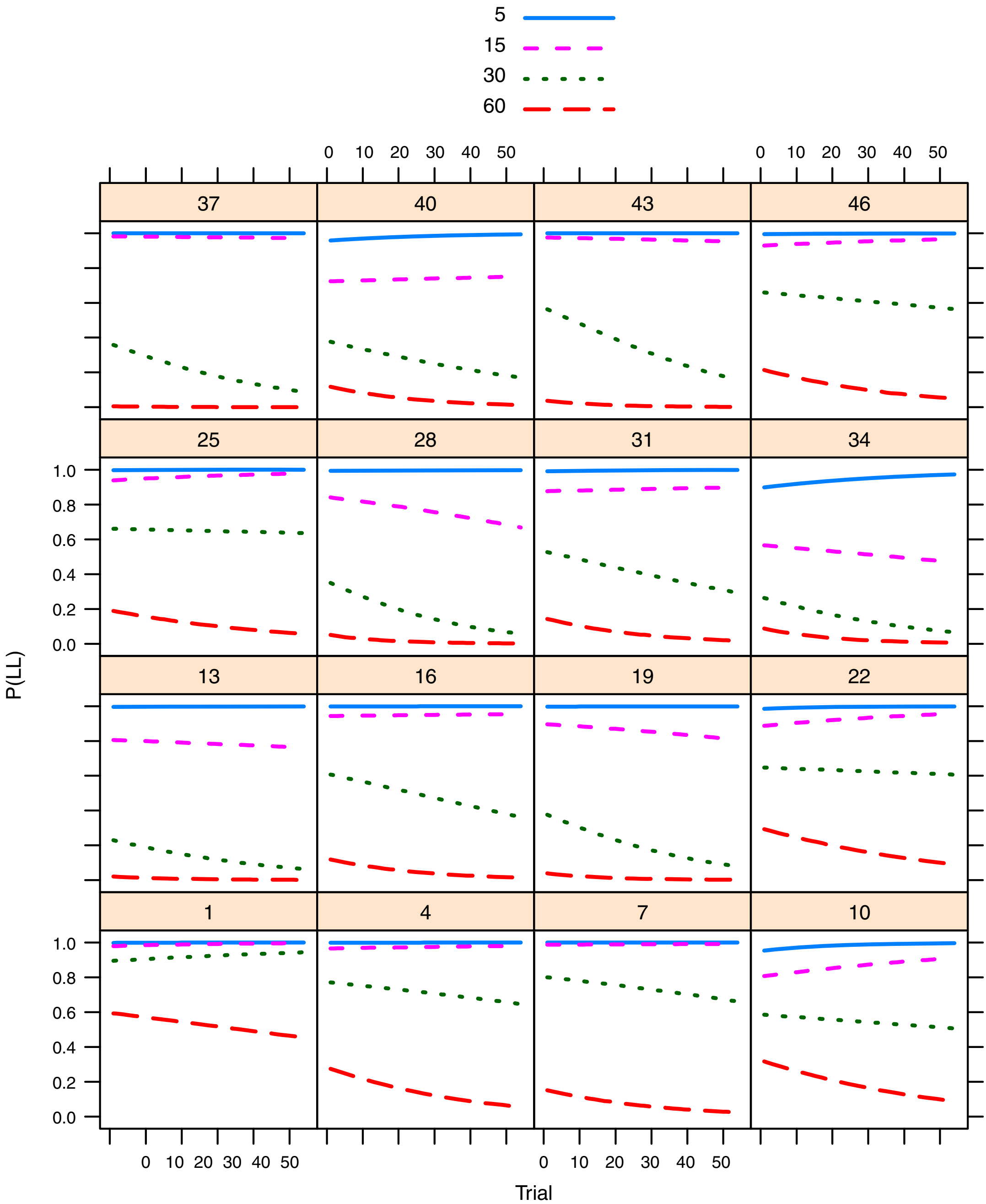

Fig. 6. Fitted probability of choosing the larger-later outcome at each delay across trials for each of the subjects in Peterson et al. (2015). Fits were derived using a multilevel logistic regression model that included trial as a moderator of the effects of magnitude and delay ratios on choice.

sensitivity to magnitude did not. Although not shown here, a similar analysis assessing changes across sessions within a block of training (which always involved the same delay ratio) revealed increases in delay sensitivity across sessions but a decrease in magnitude sensitivity (the *P(LL)* at the average delay ratio decreased from 88% to 73% over the five sessions of training).

One drawback to the design underlying this analysis is that only one magnitude ratio was tested (2:1), therefore the results may not translate to other magnitude ratios (see Fig. 4). To more fully explore the effects of magnitude differences, delay differences, and their various combinations on the relative likelihood of choosing the LL outcome over the SS outcome, a design would need to use carefully chosen combinations of ratios (McClelland & Judd, 1993).

Fitting Data with Unusual Participants

In the final analysis, I considered a dataset generated using a video game preparation in which participants could either fire their weapon quickly to obtain a SS outcome or wait to fire their weapon to obtain a LL outcome (Young & McCoy, 2015). The SS outcome was always available after 0.25 s whereas the LL outcome was available at a delay that changed on occasion during game play, varying uniformly within the 2 s to 8 s interval. The magnitude of the SS outcome was 5 points of damage and that for the LL was 100 points of damage. For simplicity, this example data set retained one of the original between-subject task variables (delay discounting vs. deferred gratification) and was limited to the final block of training.

This data set is of particular interest because some of the subjects showed quite extreme behavior. Of the 40 subjects, 3 of them chose the LL outcome 100% of the time. These are the types of behavioral patterns that are especially problematic for a two-stage analysis in which individual fits are the basis for a second analysis as well as for techniques relying on the existence of indifference points. When individual logistic regressions were fitted to each of the subjects, those who always chose the LL generated intercept estimates (magnitude sensitivity) of 2.66 with a SEs over 200,000 and delay slope estimates less than 0.0001 with SEs over 130,000, clear evidence of perfect separation. If these subjects were dropped, differential attrition would arise because all three were in the delay discounting condition.

A multilevel logistic regression was performed using the intercept to estimate the magnitude effect and effect coding of the two-level task variable (see Supplemental Materials for details). The important consequences of this analysis are that there were no problems with perfect separation, and the estimates for the three extreme subjects revealed strong sensitivity to magnitude (intercepts ~ 6.6) and weak sensitivity to delay (slopes ~ -0.9). Although the model estimated some sensitivity to delay in the absence of any supporting data from each of these subjects, the model predicted that some limited sensitivity would emerge based on the behavior of similar subjects in the same condition. As expected given their high likelihood of choosing the LL, dropping these three subjects from the multilevel analysis had a large impact on the main effect of task (*P(LL)* changed from 16% vs. 70% for the deferred gratification and delay discounting tasks, respectively, to 16% and 42%), but had little impact on the overall estimate of delay sensitivity (delay slopes changed from -1.24 and -0.56, respectively, to -1.24 and -0.48, well within the SE of 0.40 to 0.61 depending on task).

Recap and New Challenges

Logistic multilevel modeling provides a theoretically sound basis on which to estimate the effects of delay and magnitude on animal choice. It seamlessly incorporates the differential validity of each subject's data by considering behavioral consistency, sample size, and unusual data values. The method does not necessitate the elimination of individual subjects who fail an inclusion criterion that is arbitrarily decided by researchers. Rather, subjects should be eliminated based on obvious criteria like failing to follow directions, sleeping during the session, or falling ill. Although I strongly advocate for the application of multilevel logistic regression to analyze delay discounting choice data, the approach will present new challenges to the researcher.

Logistic regression. One of the biggest barriers to the adoption of a logistic regression framework for understanding impulsive choice involves the specific challenges of regression. Analysis of variance (ANOVA) is a much more familiar analytic tool, but it is not suitable for the analysis of binomial outcomes. Furthermore, despite ANOVA being a special case of regression (both are encompassed under the general linear model), regression analysis requires that a data analyst have familiarity with the effects of various methods of coding categorical variables as predictors, the likelihood of collinearity between main effects and interactions, the possibility of nonlinear relationships between a predictor and the outcome, and overfitting caused by the use of a model that is too complex given the sample size (Cohen, Cohen, West, & Aiken, 2002). The interpretation of the results of a logistic regression is further complicated by the need to understand that the resulting statistical model is no longer $y = \Sigma b_i x_i$ but now $P(y = 1) = logistic(\Sigma b_i x_i)$ with each of the regression weights, b_i, now predicting changes in the log odds of the outcome.

Logistic regression is a special case of the generalized linear model that allows the specification of error distributions other than the normal/Gaussian distribution. The machinery that underlies these models can produce errors unfamiliar to the conventionally trained data analyst. One such error is a "failure to converge" which can be caused by complete separation between the two classes of outcomes being distinguished. Although there are solutions for this problem, when encountered the message will initially vex the researcher. Furthermore, logistic regression models should be compared using likelihood-based metrics like the likelihood ratio test, AIC or BIC due to the issues with pseudo-R^2 indices (Cohen et al., 2002). Finally, the standard diagnostic tests familiar to users of regression (leverage, influence, linearity) must be adapted and interpreted more cautiously in logistic regression.

Multilevel analysis. Another significant challenge to adopting multilevel logistic regression as the preferred tool is the need to understand multilevel modeling. Generalized multilevel modeling provides considerable utility to the researcher who needs to assess categorical and continuous predictors within a repeated measures design, and it is the only suitable method when the outcome variable is binomial. Furthermore, multilevel modeling of repeated measures data provides a superior method for handling missing data and unbalanced designs (Everitt, 1998; Gueorguieva & Krystal, 2004). Given the dramatic increase in the use of multilevel modeling over the past few years (Boisgontier & Cheval, 2016), all researchers conducting repeated measures designs are encouraged to gain familiarity with the technique.

Treatment of the intercept. In a logistic regression of a delay discounting experiment, the probability of choosing the LL response is 50% when the prediction equation produces a zero. This situation corresponds to predicting indifference when choosing between the SS and LL options. For Equation (2), the prediction equation predicts indifference when the magnitude ratio is 1 (no difference in magnitudes between the two options) and the LL delay is zero (thus equal to the SS delay). In a well-designed experiment, this indifference is precisely what should occur if the choice options are appropriately counterbalanced. Thus, the inclusion of an intercept is not necessary to capture a bias toward one option versus the other when the SS and LL are identical.

However, if the magnitude ratio or delay ratio is held constant throughout the experiment (e.g., Peterson et al., 2015), then it is acceptable to include the intercept to capture the impact of the constant nonzero ratio. If the intercept is used for this purpose, it is critical that the other predictors have a value of zero when there should be no bias, $P(LL) = 0.5$, so that deviations from that value indicates an effect of the factor that was held constant. Thus, log(ratios) should not be centered as commonly done in regression analyses, but other predictors like trial, session, or categorical predictors can be centered if the goal is to estimate the effects of magnitude and delay ratios at the average value of the other predictors.

The decision to include the intercept to capture the effects of a factor that was held constant alters the assessment of the regression weights involving that factor. For example, if the dataset included a column corresponding to a 2:1 LL:SS magnitude ratio which was then used as a predictor in the regression (while omitting the intercept), then the regression weight for this predictor would be estimating $\beta_{magnitude}$ in Equation (3). However, if this constant factor was estimated by simply including the intercept, then β_0 is estimating $\beta_{magnitude} \times \log(\mathrm{LL_{magnitude}}/\mathrm{SS_{magnitude}})$ requiring that the constant log(ratio) would need to be factored out to obtain a direct estimate of the regression weight (e.g., $\beta_{magnitude} = \beta_0 / log(2)$ for a 2:1 constant magnitude ratio). The same scaling issue would need to be considered when estimating any interactions relevant to the constant factor. Of course, if the researcher's goal is only to identify that the regression weight was different from zero or that it was different across conditions or groups, then the scaling issue can be ignored.

Design. Some common delay discounting designs are not suited to assessing the separate influence of magnitude and delay ratios because these variables are highly correlated in that design (e.g., Kirby et al., 1999), but this type of design is well-suited to sampling along the diagonal of the right side of Figure 2 or the bottom of Figure 5 in order to distinguish *k* values. In other designs, only one level of a variable is assessed (e.g., one magnitude ratio), thus raising the question of whether the effects of the other predictors are specific to that particular level. Consequently, researchers interested in reaching broader conclusions about the effects of magnitude and delay ratios on the choice between SS and LL outcomes should design their studies to broadly sample a range of ratios for each factor (although carryover and learning effects must be considered). Additionally, given the logarithmic effect of delay and magnitude ratios on behavior, smaller ratios need to be more heavily sampled than larger ratios.

It is easiest to conceptualize such a design in terms of a full factorial combination of each factor sampled along a logarithmic scale. The left side of Figure 7 illustrates such a design for a study in which the magnitude ratio and delay ratios vary from 1 to 100. Subjects would be assessed at each of these combinations one or more times to identify the effect of each variable on behavior. An alternative more highly powered design would only sample the four corners of this space (McClelland & Judd, 1993), but such a design would limit the ability to identify individual differences. A final proposal is to use Halton sequences (Halton, 1964) in which the parameter space is gradually filled as more combinations are tested. This approach starts by sampling a few widely spaced points in the parameter space, and with each additional sampling it identifies a combination that lies in the interstices of the parameter space. The middle and right sides of Figure 7 show a Halton sequence generating 25 combinations versus 50. The primary advantage of a Halton sequence is its sampling efficiency when more than two variables must be sampled as well as its effectiveness when fewer or more combinations are tested than were originally planned. It is important to note that all of these approaches assume that a particular ratio is equivalent across the range of magnitudes and delays. If the researcher suspects that a 2:1 ratio is different from a 100:50 ratio, then it would be necessary to also manipulate the basis of the ratio.

Conclusion

The decomposition of *k* into its components of reward and delay sensitivity creates opportunities for delay discounting research. Impulsivity has been increasingly considered a multifaceted construct because various measures of impulsivity are differentially correlated and load on different factors in a factor analysis (Dougherty et al., 2009; Evenden, 1999;

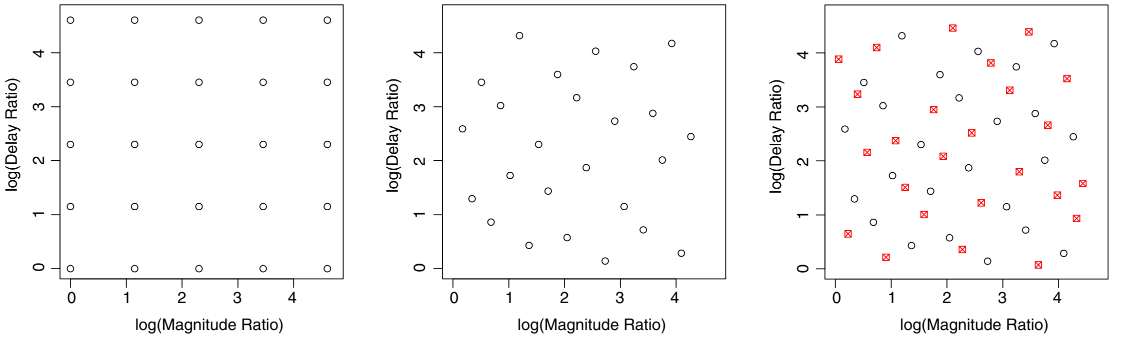

Fig. 7. Methods of sampling two variables to maximize parameter range coverage. (Left) Factorial design (5 × 5) of 25 parameter combinations. (Middle) Halton sequence (base 2 and 3) of 25 parameter combinations. (Right) Halton sequence (base 2 and 3) of 50 parameter combinations (additional 25 combinations are indicated with a new plot symbol).

Perry & Carroll, 2008; Robinson et al., 2009). Thus, the decomposition of the delay discounting rate into two slopes, one capturing differential sensitivity to the reward ratio and the other capturing differential sensitivity to the delay ratio, is consistent with this finding, and decomposing *k* should provide opportunities for a greater understanding of the factors affecting the choice between SS and LL outcomes.

By fitting a multilevel logistic regression model, the researcher employs a framework that can evaluate various models of the discounting process. The relationships between magnitude and delay ratio and choice could be linear, logarithmic, power, or another function of each predictor, and the model can test for interactions among the predictors. The rationale for the assumption of a logarithmic relationship in the current treatise is based on psychophysical experiments (Buhusi & Meck, 2005; Fechner, 1860; Gibbon, 1977). The log transform of the ratio between two magnitudes is equal to the difference between the individual log transforms of the magnitudes: $\log(LL_{magnitude}/SS_{magnitude}) = \log(LL_{magnitude}) - \log(SS_{magnitude})$. Furthermore, Cui (2011) noted that hyperbolic discounting emerges when both the perception of value and time are assumed to follow Weber's law (a logarithmic relationship). Regardless, various functional relationships between magnitude, delay and choice have been proposed (for a review, see Doyle, 2013) and can be tested using the multilevel logistic approach.

Furthermore, I have demonstrated that a multilevel logistic regression evaluates behavior at the trial level rather than aggregating across trials to identify indifference points. This approach makes it much easier to examine changes in behavior across trials (e.g., Fig. 6). This analytical approach also would permit the evaluation of models that assume that the choice on trial n is a function of the choice or latency on trial $n - 1$, $n - 2$, etc. These types of questions are not addressable by fitting indifference point data.

Finally, the advocated logistic approach can be readily extended to include options that differ in their relative probability of producing a reward. Not only could probability discounting (Kahneman & Tversky, 1979; Rachlin, 2006; Rachlin et al., 1991) be analyzed using a ratio-based logistic model, but designs incorporating choices between options that differ in any combination of magnitude, delay, and probability can be similarly analyzed. The log(probability ratio), however, may not be the best predictor; instead, a log(odds ratio) may be more suitable (Rachlin et al., 1991). Of course, any such analysis depends on good design, and experiments in which there are insufficient combinations of magnitudes, delays, and probabilities would limit the ability to assess their independent impact or to test interactions among these factors (McClelland & Judd, 1993). Regardless, a logistic approach to analyzing choice generalizes to a wide range of experimental designs and would not be limited to situations in which the discounting rates have been identified for a discrete set of predetermined choices (Gray et al., 2016; Kirby et al., 1999).

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